From fusion rules to solving the pentagon identity

This notebook starts from the (multiplicity-free) fusion rules shared by the Ising and $SU(2)_2$ fusion rings. It creates every admissible *F*-symbol, fixes a simple real gauge, generates (all components of) the pentagon identity, solves the full system, and extracts the $F\,Z\,F = X$ block.

References

• Simon's book

• Bonderson's thesis: https://thesis.caltech.edu/2447

Bonderson uses Mathematica, but gives no code. More references at the end of this notebook.

Fusion rules

The labels 1, 2, 3 correspond to the three simple sectors. The only input here is the fusion table.

In[•]:=
```
Remove["Global`*"] // Quiet
labels = {1, 2, 3};
fusion[1, a_] := {a}; fusion[a_, 1] := {a};
fusion[2, 2] = {1, 3}; fusion[2, 3] = {2};
fusion[3, 2] = {2}; fusion[3, 3] = {1};
allowedQ[a_, b_, c_] := MemberQ[fusion[a, b], c];
```

A quick sanity check: 2 x 2 has the two channels 1 and 3.

In[•]:=
```
fusion[2, 2]
```

Out[•]=

$\{1, 3\}$

Admissible F-symbols

Write `f[a,b,c,d,e,f]` = $(F_d^{abc})_{ef}$. It relates the combination $((a\ b)_e\ c)_d$ to the combination $(a\ (b\ c)_f\ f)_d$. An entry makes sense (is admissible) only when both fusion trees are allowed.

In[•]:=
```
admissibleFQ[a_, b_, c_, d_, e_, f_] :=
  allowedQ[a, b, e] ∧ allowedQ[e, c, d] ∧ allowedQ[b, c, f] ∧ allowedQ[a, f, d];
fName[a_, b_, c_, d_, e_, f_] :=
  Symbol["f" <> StringJoin[Map[ToString, {a, b, c, d, e, f}]]];
```

First fix the standard unit gauge: inadmissible entries are zero, while an associator involving the vacuum 1 is set to 1. All other admissible entries remain independent real unknowns.

In[•]:=
```
Fsymbol[a_, b_, c_, d_, e_, f_] :=
  Which[! admissibleFQ[a, b, c, d, e, f],
   0, MemberQ[{a, b, c}, 1], 1, True, fName[a, b, c, d, e, f]];
```

(Here the symbol "!" stands for "not")

Enumerate all remaining unknown F-symbols:

In[•]:=
```
admissibleNonUnitTupleQ[t_List] :=
 admissibleFQ[t[[1]], t[[2]], t[[3]], t[[4]], t[[5]], t[[6]] ∧
  FreeQ[Take[t, 3], 1]
fVariableFromTuple[t_List] :=
 fName[t[[1]], t[[2]], t[[3]], t[[4]], t[[5]], t[[6]]]
unknownTuples = Select[Tuples[labels, 6], admissibleNonUnitTupleQ]
unknowns = Map[fVariableFromTuple, unknownTuples]
Length[unknowns]
```

Out[•]=

$$\begin{pmatrix}
2 & 2 & 2 & 2 & 1 & 1 \\
2 & 2 & 2 & 2 & 1 & 3 \\
2 & 2 & 2 & 2 & 3 & 1 \\
2 & 2 & 2 & 2 & 3 & 3 \\
2 & 2 & 3 & 1 & 3 & 2 \\
2 & 2 & 3 & 3 & 1 & 2 \\
2 & 3 & 2 & 1 & 2 & 2 \\
2 & 3 & 2 & 3 & 2 & 2 \\
2 & 3 & 3 & 2 & 2 & 1 \\
3 & 2 & 2 & 1 & 2 & 3 \\
3 & 2 & 2 & 3 & 2 & 1 \\
3 & 2 & 3 & 2 & 2 & 2 \\
3 & 3 & 2 & 2 & 1 & 2 \\
3 & 3 & 3 & 3 & 1 & 1
\end{pmatrix}$$

Out[•]=
```
{f222211, f222213, f222231, f222233, f223132, f223312,
  f232122, f232322, f233221, f322123, f322321, f323222, f332212, f333311}
```

Out[•]=
```
14
```

So, for this fusion ring, there are 14 unknowns, after picking unit gauge.

Generate the complete pentagon system

The initial tree is (((a b)_p c)_q d)_t and the final tree is (a (b (c d)_s)_r)_t. For every admissible choice of labels we equate the three-F “long route” with the two-F “short route”.

In[•]:=
```
pentagonTupleQ[{a_, b_, c_, d_, t_, p_, q_, r_, s_}] :=
  allowedQ[a, b, p] ∧ allowedQ[p, c, q] ∧ allowedQ[q, d, t] ∧
   allowedQ[c, d, s] ∧ allowedQ[b, s, r] ∧ allowedQ[a, r, t];
pentagonEquation[{a_, b_, c_, d_, t_, p_, q_, r_, s_}] :=
  Module[{x, longRoute, shortRoute},
   longRoute =
    Total[Table[Fsymbol[a, b, c, q, p, x] ×
        Fsymbol[a, x, d, t, q, r] × Fsymbol[b, c, d, r, x, s], {x, labels}]];
   shortRoute =
    Fsymbol[p, c, d, t, q, s] × Fsymbol[a, b, s, t, p, r];
   longRoute == shortRoute];
```

Find admissible labelings of the pentagon. After deleting identities and duplicate equations, nontrivial polynomial equations remain.

In[•]:=
```
pentagonTuples = Select[Tuples[labels, 9], pentagonTupleQ];
pentagonEquations =
 DeleteDuplicates[DeleteCases[Map[pentagonEquation, pentagonTuples],
   True]]
{Length[pentagonTuples], Length[pentagonEquations]}
```

Out[•]=

$$\{f222211^2 + f222213\, f222231\, f232122 = 1,\ f222211\, f222213 + f222213\, f222233\, f232122 = 0,$$
$$f222211\, f222231 + f222231\, f222233\, f232122 = 0,\ f222213\, f222231 + f222233^2\, f232122 = f223132\, f322123,$$
$$f222211^2 + f222213\, f222231\, f232322 = 0,\ f222211\, f222213 + f222213\, f222233\, f232322 = f223312,$$
$$f222211\, f222231 + f222231\, f222233\, f232322 = f322321,\ f222213\, f222231 + f222233^2\, f232322 = 0,$$
$$f222213\, f223132\, f233221 = f222211,\ f222211\, f223312 = f222213,\ f222233\, f223132\, f233221 = f222231\, f323222,$$
$$f222231\, f223312 = f222233\, f323222,\ f222231\, f223312\, f232122 = f222211,\ f222233\, f223312\, f232322 = f222213,$$
$$f222211\, f223132\, f232122 = f222231\, f332212,\ f222213\, f223132\, f232322 = f222233\, f332212,$$
$$f223132\, f223312\, f233221 = 1,\ f223132\, f223312\, f233221 = f333311,\ f222211\, f232122\, f322123 = f222213\, f233221,$$
$$f222213\, f232122\, f322321 = f222211,\ f222231\, f232322\, f322123 = f222233\, f233221,$$
$$f222233\, f232322\, f322321 = f222231,\ f223132\, f232322\, f323222 = f223132\, f232122,$$
$$f223312\, f232122\, f323222 = f223312\, f232322,\ f233221\, f332212 = f232122^2,\ f233221\, f332212 = f232322^2,$$
$$f233221\, f333311 = f233221,\ f222231\, f322123\, f332212 = f222211,\ f222233\, f322123\, f332212 = f222213\, f323222,$$
$$f222211\, f322321 = f222231,\ f222213\, f322321 = f222233\, f323222,\ f223312\, f322321 = f223132\, f322123,$$
$$f223132\, f322123\, f333311 = f223312\, f322321,\ f232322\, f322123\, f323222 = f232122\, f322123,$$
$$f232122\, f322321\, f323222 = f232322\, f322321,\ f233221\, f323222^2 = f233221,\ f322123\, f322321\, f332212 = 1,$$
$$f322123\, f322321\, f332212 = f333311,\ f323222^2\, f332212 = f332212,\ f332212\, f333311 = f332212,\ f333311^2 = 1\}$$

Out[•]=

$$\{136, 41\}$$

So there are 136 labelings and 41 equations.

Fix the residual real gauge

The one-dimensional fusion vertices still have basis freedom. We choose three convenient real conventions below. These are *gauge choices*, not consequences of the pentagon.

In[•]:=
```
gaugeEquations = {
    fName[2, 3, 2, 3, 2, 2] == -1,
    fName[3, 2, 2, 3, 2, 1] == 1,
    fName[3, 3, 2, 2, 1, 2] == 1
   };
```

Solve all pentagons

Solve the full polynomial system over the reals. In this gauge Mathematica finds two discrete coherent solutions.

Display all 14 solved entries for either branch. The two branches differ in the sign of the nontrivial F^{222}_2 block.

In[•]:=
```
solutions =
 FullSimplify[Solve[Join[pentagonEquations, gaugeEquations], unknowns, Reals]]
Length[solutions]
```

Out[•]=

$$\left\{\left\{\text{f222211} \to -\frac{1}{\sqrt{2}}, \text{f222213} \to -\frac{1}{\sqrt{2}}, \text{f222231} \to -\frac{1}{\sqrt{2}}, \text{f222233} \to \frac{1}{\sqrt{2}}, \text{f223132} \to 1, \text{f223312} \to 1, \text{f232122} \to 1,\right.\right.$$
$$\left.\left.\text{f232322} \to -1, \text{f233221} \to 1, \text{f322123} \to 1, \text{f322321} \to 1, \text{f323222} \to -1, \text{f332212} \to 1, \text{f333311} \to 1\right\},\right.$$
$$\left\{\text{f222211} \to \frac{1}{\sqrt{2}}, \text{f222213} \to \frac{1}{\sqrt{2}}, \text{f222231} \to \frac{1}{\sqrt{2}}, \text{f222233} \to -\frac{1}{\sqrt{2}}, \text{f223132} \to 1, \text{f223312} \to 1, \text{f232122} \to 1,\right.$$
$$\left.\left.\text{f232322} \to -1, \text{f233221} \to 1, \text{f322123} \to 1, \text{f322321} \to 1, \text{f323222} \to -1, \text{f332212} \to 1, \text{f333311} \to 1\right\}\right\}$$

Out[•]=

2

Note these are **two solutions**.

For compactness, pick solution #2 (but feel free to change it below):

In[•]:=
```
solutionTable[sol_List] :=
  Table[
   {unknownTuples[[n]], FullSimplify[unknowns[[n]] /. sol]},
   {n, Length[unknowns]}
  ];
Grid[Prepend[solutionTable[solutions[[2]]],
  {"{a,b,c,d,e,f}", DisplayForm@SubscriptBox[
     RowBox[{"(", SubsuperscriptBox["F", "d", RowBox[{"a", "b", "c"}]], ")"}],
     RowBox[{"e", "f"}]]}], Frame → All]
```

Out[•]=

$\{a,b,c,d,e,f\}$	$\left(F_d^{a\,b\,c}\right)_{e\,f}$
{2, 2, 2, 2, 1, 1}	$\frac{1}{\sqrt{2}}$
{2, 2, 2, 2, 1, 3}	$\frac{1}{\sqrt{2}}$
{2, 2, 2, 2, 3, 1}	$\frac{1}{\sqrt{2}}$
{2, 2, 2, 2, 3, 3}	$-\frac{1}{\sqrt{2}}$
{2, 2, 3, 1, 3, 2}	1
{2, 2, 3, 3, 1, 2}	1
{2, 3, 2, 1, 2, 2}	1
{2, 3, 2, 3, 2, 2}	−1
{2, 3, 3, 2, 2, 1}	1
{3, 2, 2, 1, 2, 3}	1
{3, 2, 2, 3, 2, 1}	1
{3, 2, 3, 2, 2, 2}	−1
{3, 3, 2, 2, 1, 2}	1
{3, 3, 3, 3, 1, 1}	1

Pick the branch with positive upper-left entry in F^{222}_2:

In[•]:=
```
sol = solutions[[2]]
```

Out[•]=

$$\left\{\text{f222211} \to \frac{1}{\sqrt{2}}, \text{f222213} \to \frac{1}{\sqrt{2}}, \text{f222231} \to \frac{1}{\sqrt{2}}, \text{f222233} \to -\frac{1}{\sqrt{2}}, \text{f223132} \to 1, \text{f223312} \to 1, \text{f232122} \to 1,\right.$$
$$\left.\text{f232322} \to -1, \text{f233221} \to 1, \text{f322123} \to 1, \text{f322321} \to 1, \text{f323222} \to -1, \text{f332212} \to 1, \text{f333311} \to 1\right\}$$

Now assemble the four scalar entries of F^{222}_2, with intermediate channels ordered as {1,3}.

In[•]:=
```
channels = {1, 3};
F =
 FullSimplify[Table[Fsymbol[2, 2, 2, 2, e, f], {e, channels}, {f, channels}] /. sol]
```

Out[•]=

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Extract one four-particle pentagon: F Z F = X

Specialize to four external 2's with total channel 3. The middle long-route recoupling is diagonal in the intermediate channel.

In[•]:=
```
Z =
 FullSimplify[
  DiagonalMatrix[Table[Fsymbol[2, x, 2, 3, 2, 2], {x, channels}]] /. sol]
```

Out[•]=

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The complete short route is obtained from the two scalar F-symbols on that side of the pentagon.

In[•]:=
```
X = FullSimplify[Table[Fsymbol[p, 2, 2, 3, 2, s] × Fsymbol[2, 2, s, 3, p, 2],
     {p, channels}, {s, channels}] /. sol]
```

Out[•]=

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Finally check the selected component of the pentagon. It is not an extra assumption: it is one small block extracted from the full solved pentagon system.

In[•]:=
```
FullSimplify[F.Z.F == X]
```

Out[•]=

True

Expected result: *F* is the Hadamard matrix, *Z* = diag(1,-1), *X* swaps the two channels, and the last command returns **True**. The fusion ring supplies the allowed vertices, while the pentagon constrains the additional recoupling data.

More “advanced” references:

- *Aboumrad* (SageMath) https://arxiv.org/abs/2212.00831
- *Anyonica* (Mathematica package)